

ALI TURAB

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EDUCATION

COMSATS University Islamabad

Lahore

Bachelors in Software Engineering

Expected Graduation, June 2026

- CGPA: 3.69

EXPERIENCE

NETSOL Technologies Pakistan

Lahore

AI/ML Intern

July 2025 – August 2025

- Developed a **Finance** App integrating **Deep Neural Networks** and **RAG** architecture to help users manage their finances.
- Designed and developed **responsive UI** components in **React + TypeScript**, with a **FastAPI** backend.
- Implemented secure user authentication using **OAuth 2.0**, including **Google Sign-In** integration.
- Optimized frontend performance, component **re-render efficiency** and improved **UX flow** by simplifying navigation.

AIESEC in Pakistan

Lahore

Team Member (Business Development, Out-Going Volunteer)

Sep 2023 - Sep 2024

- Built **partnerships** and managed stakeholders while maintaining global AIESEC relations.
- Applied **problem-solving** and **adaptability** to support community and business development initiatives.

PROJECTS

Shariah Law Assistant

Personal Project

July 2025

- Curated and preprocessed texts into structured **datasets** for **multilingual embeddings**.
- Developed a **RAG** pipeline using **LangChain**, **ChromaDB**, and **Gemini-1.5-Flash**, enabling context-aware Q&A.
- Deployed backend in **FastAPI**, integrating **paraphrase-multilingual-mpnet-base-v2** to support Arabic queries.

FutureFinance

Internship Project

July 2025 - August 2025

- Implemented a **RAG** pipeline (**LangChain + OpenAI API**) for personalized saving strategies using curated synthetic data of 5,000+ local products.
- Built investment module by training an **LSTM** model on **10 years** of Vanguard S&P 500 ETF data, achieving a **Mean Squared Error of 5.13** for trend prediction. Developed a **clean insights dashboard** to display it all.
- Created stock forecasting and insights using **linear regression + gradient descent**, and integrated live financial news and market updates via **APIs** while focusing on **user-centric layout decisions**

NavVision – Helping Visually Impaired People

Final Year Project

July 2025 – Current

- Built using **YOLOv8n (104 classes, 20k+ images)**, **ZoeD-M12-NK**, and **BLIP** on a **Flutter + Python** stack.
- Designed async inference (**YOLO ~15FPS**, **ZoeDepth ~1FPS**) with **FP16 + 640×480** optimization for edge devices
- Delivered depth-based alerts + scene captions; developing a **Smart Voice Agent** (in progress).

SKILLS / TOOLS

- **Programming Languages:** Java, Springboot, C, C++, JavaScript, TypeScript, Python.
- **Frontend Development:** HTML, CSS, React.js, Next.js, Tailwind CSS, Bootstrap, Material UI, Figma, Framer Motion
- **Backend Development:** Node.js, Express.js, FastAPI, REST APIs
- **State Management:** Redux, Redux Toolkit.
- **Databases:** PostgreSQL, MongoDB, Supabase, MS SQL Server.
- **AI/ML:** TensorFlow, PyTorch, LangChain, LangGraph, OpenAI, Hugging Face, OpenCV.
- **Tools:** GitHub, Docker, Streamlit, Stripe